

ACM CODE OF ETHICS AND PROFESSIONAL CONDUCT

GUIDING PRINCIPLES FOR COMPUTING PROFESSIONALS

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INTRODUCTION

- What is the ACM?
- Importance of Ethics in Computing
- Purpose of the ACM Code of Ethics

STRUCTURE OF THE CODE

- General Ethical Principles
- Professional Responsibilities
- Leadership Responsibilities
- Compliance with the Code

PRINCIPLE 1.1 – CONTRIBUTE TO SOCIETY AND TO HUMAN WELL-BEING

- Promote human rights and environmental sustainability
- Minimize negative consequences like privacy violations
- Engage in pro bono or volunteer work



PRINCIPLE 1.2 – AVOID HARM

- Identify and mitigate harms
- Report risks responsibly
- Use best practices and ethical justification for any intentional harm

PRINCIPLE 1.3 – BE HONEST AND TRUSTWORTHY

- Provide full disclosure of system capabilities and limitations
- Avoid deception, honor commitments
- Be transparent about qualifications

PRINCIPLE 1.4 – BE FAIR AND TAKE ACTION NOT TO DISCRIMINATE

- Avoid prejudicial discrimination
- Design inclusive and accessible systems
- Promote fairness in decision-making



PRINCIPLE 1.5 – RESPECT INTELLECTUAL PROPERTY

- Credit original creators
- Respect legal protections like copyrights and patents
- Support open-source and public domain contributions



PRINCIPLE 1.6 – RESPECT PRIVACY

- Use data only for legitimate ends
- Ensure transparency and informed consent
- Minimize data collection and retention

PRINCIPLE 1.7 – HONOR CONFIDENTIALITY

- Protect sensitive information
- Disclose only to appropriate authorities when necessary

PRINCIPLE 2.1 – HIGH QUALITY IN WORK

- Insist on high standards in processes and products
- Respect dignity and transparent communication
- Avoid inducements to compromise quality



PRINCIPLE 2.2 – MAINTAIN PROFESSIONAL COMPETENCE

- Stay current with skills and knowledge
- Engage in continuous learning and development
- Encourage professional growth in teams



PRINCIPLE 2.3 – KNOW AND RESPECT LAWS

- Follow legal and organizational policies
- Challenge unethical rules responsibly
- Accept consequences of ethical decisions



PRINCIPLE 2.4 – ACCEPT AND PROVIDE REVIEW

- Participate in peer and stakeholder review
- Provide constructive feedback
- Improve quality through collaboration



PRINCIPLE 2.5 – EVALUATE SYSTEMS AND RISKS

- Analyze potential consequences
- Report major risks
- Avoid deployment of unsafe systems

PRINCIPLE 2.6 – WORK WITHIN COMPETENCE

- Evaluate assignments for feasibility
- Disclose limits of expertise
- Acquire skills or defer tasks responsibly

PRINCIPLE 2.7 – FOSTER PUBLIC UNDERSTANDING

- Educate others about computing impacts
- Address misinformation respectfully
- Promote responsible use of technology

PRINCIPLE 2.8 – ACCESS RESOURCES ETHICALLY

- Avoid unauthorized access
- Act only under public good with care
- Mitigate harm in exceptional cases



PRINCIPLE 2.9 – BUILD SECURE SYSTEMS

- Design for robust and usable security
- Monitor and patch vulnerabilities
- Inform stakeholders about breaches

LEADERSHIP RESPONSIBILITY – PUBLIC GOOD AS CENTRAL CONCERN

- Consider impact on all stakeholders
- Apply ethics in every project phase
- Use ethical methodologies and design

LEADERSHIP – ENCOURAGE ETHICAL CONDUCT

- Foster social responsibility
- Promote full participation
- Discourage unethical behavior

LEADERSHIP – ENHANCE QUALITY OF WORK LIFE

- Support well-being and safety
- Respect ergonomic and accessibility needs
- Provide growth opportunities



LEADERSHIP – ETHICAL POLICY AND PROCESSES

- Define and communicate ethical policies
- Reward compliance, address violations
- Avoid enabling unethical practices

LEADERSHIP – SUPPORT LIFELONG LEARNING

- Provide education on ethics and professionalism
- Familiarize staff with system impacts
- Encourage skill and judgment development



LEADERSHIP – MANAGING SYSTEM CHANGES

- Plan changes carefully
- Avoid harmful deprecation
- Support migration to alternatives

LEADERSHIP – STEWARDSHIP OF INTEGRATED SYSTEMS

- Monitor societal integration
- Ensure fair system access
- Develop new care standards as needed

COMPLIANCE – UPHOLD AND PROMOTE THE CODE

- Act consistently with ethical principles
- Encourage ethical behavior in others
- Report violations appropriately

COMPLIANCE – ACM MEMBERSHIP RESPONSIBILITY

- Support Code adherence across the profession
- Report breaches to ACM if needed
- Recognize consequences of unethical conduct

CASE STUDY – MALWARE DISRUPTION

- Justified harm vs. unintended consequences
- Ethical obligations to minimize risk
- Example of Principle 1.2 and 2.8

CASE STUDY – LINKING PUBLIC DATA SETS

- Re-identification risk from data aggregation
- Privacy concerns and ERB approval
- Principles 1.6, 2.4, and 2.5 involved

CASE STUDY – MEDICAL IMPLANT SECURITY

- Public good and proactive risk analysis
- Use of cryptography and open disclosures
- Principles 1.1, 2.9, 3.1, and 3.7 emphasized

CASE STUDY – ABUSIVE WORKPLACE BEHAVIOR

- Discrimination, abuse, and retaliation
- Leadership's ethical responsibility
- Violations of Principles 1.4, 3.3, and 3.4

CASE STUDY – BIASED CONTENT FILTERING

- Machine learning misused by activists
- Suppression of legitimate information
- Principles 1.2, 1.4, 2.5, and 3.7 relevant

USING THE CODE IN EDUCATION

- Integrate ethics into technical courses
- Use CARE framework and scenarios
- Promote critical thinking and awareness

USING THE CODE IN ORGANIZATIONS

- Adopt the Code in company policies
- Reward ethical behavior
- Encourage interdisciplinary collaboration



CONCLUSION

- The Code supports technical and ethical excellence
- Follow principles to benefit society and profession
- Uphold accountability and transparency in computing

REFERENCES

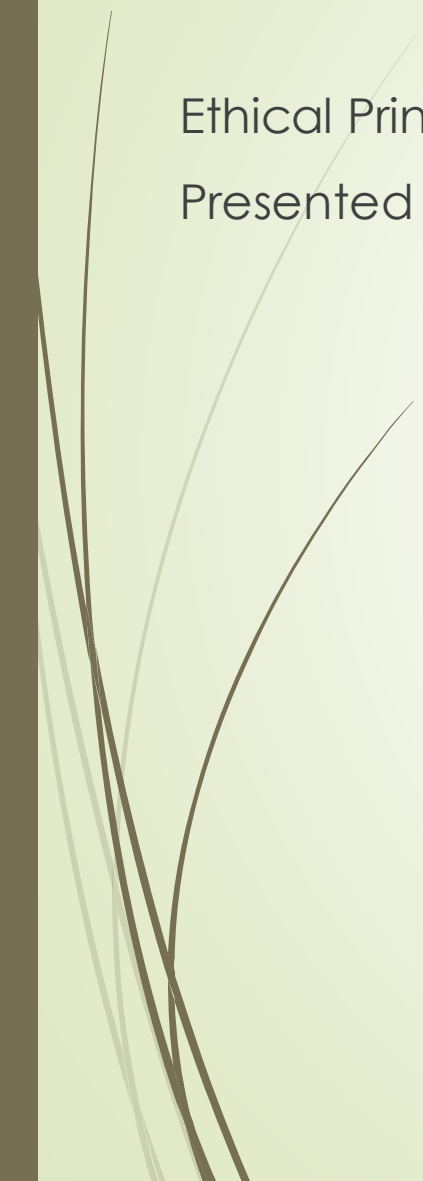
- ACM Code of Ethics and Professional Conduct (2018)
- <https://www.acm.org/code-of-ethics>



IEEE Code of Ethics

Ethical Principles for Engineers and Technologists

Presented by: Dr. Razib Hayat Khan





Introduction

What is IEEE?

Institute of Electrical and Electronics Engineers

Why ethics matter in engineering?

- Technology affects quality of life
- Responsibility towards society



Overview of the Code

Adopted: June 2020

Structure:

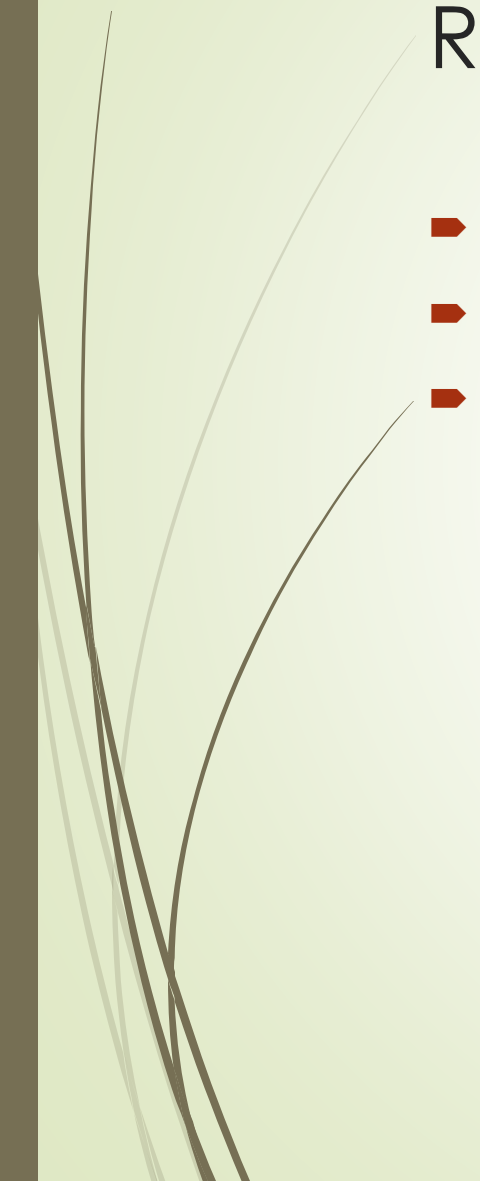
- 3 Core Ethical Duties
- 10 Guiding Principles

Purpose:

- Guide ethical behavior in professional activities



Section I – Integrity & Responsible Conduct

- Core duty: Uphold highest standards of integrity
 - Applies to all professional activities
 - Includes safety, privacy, and sustainability
- 



Principle 1

- Hold paramount: Public safety, health, welfare
- Comply with: Ethical design, sustainable development
- Disclose: Risks to public/environment
-  Discussion: Can you think of a case where failing to disclose a risk caused harm?



Principle 2

- Promote public understanding of technology
- Discuss societal implications of emerging tech (AI, robotics)
-  Example: How should we educate society about AI biases?



Principle 3

- Avoid conflicts of interest
- If unavoidable, disclose them
-  Case study: A developer recommending tools they profit from.



Principle 4

- Avoid unlawful conduct
- Reject bribery in all forms
-  Example: Vendor bribes in procurement – what to do?

Principle 5

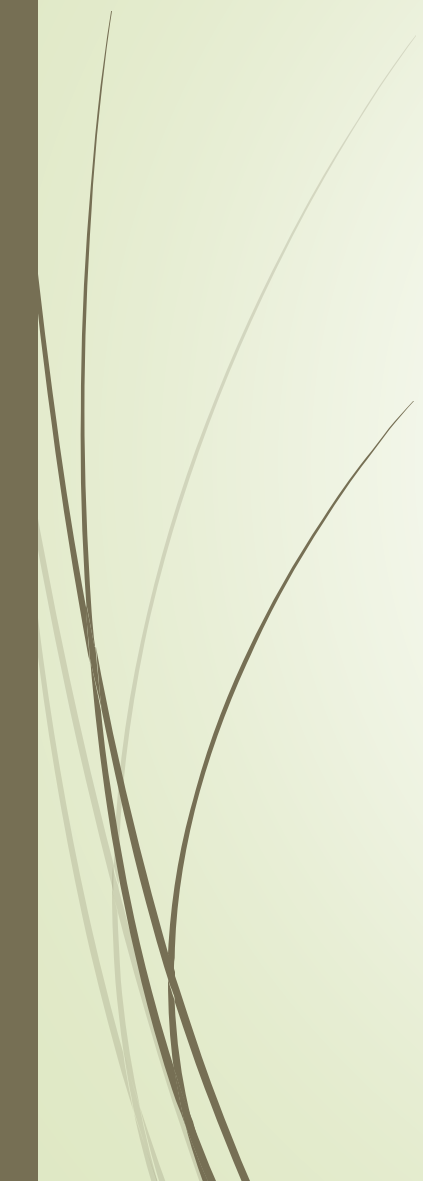
- Accept and offer honest criticism
- Correct errors, be realistic, credit others
-  Example: Plagiarism in technical papers

Principle 6

- Maintain technical competence
- Only undertake tasks if qualified
-  Ethical risk of faking expertise or overpromising



Section II – Respect & Fairness

- Treat all people with fairness and respect
 - Avoid harassment, discrimination, and harm
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Principle 7

- No discrimination by race, religion, gender, etc.
- Respect diversity and inclusion
-  Group activity: Recognizing unconscious bias



Principle 8

- No harassment or bullying
- Includes sexual harassment
-  Case scenario: Handling misconduct in teams

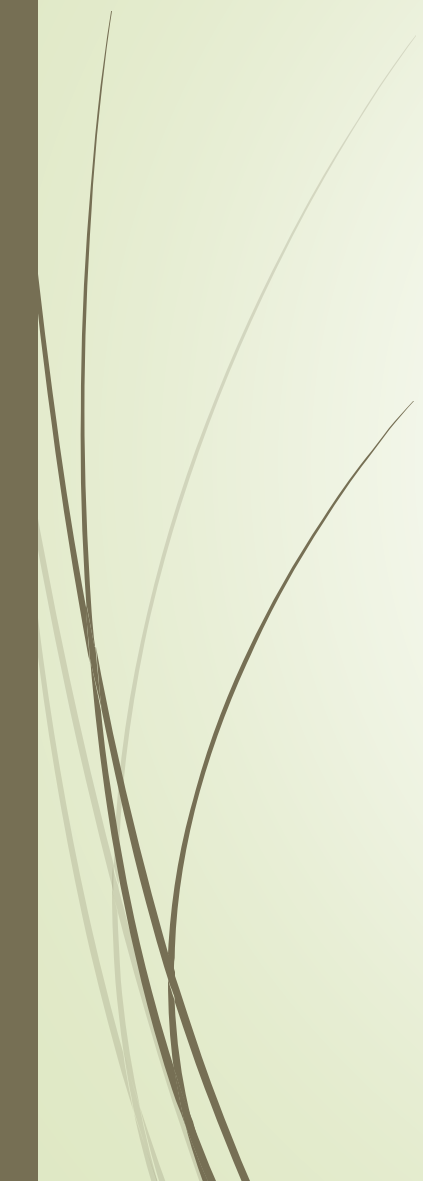


Principle 9

- Don't injure others' reputation, employment, or property
- Avoid false claims, rumors, or verbal abuse
-  Social media and professional reputation – boundaries?



Section III – Upholding the Code

- Ethical responsibility to support the code
 - Promote it among peers
 - Do not retaliate against whistleblowers
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Principle 10

- Encourage ethical behavior in peers
- Report violations
- Protect those who report unethical conduct
-  Whistleblower protection and courage to speak up



Summary of Key Principles

- Safety, honesty, respect
- Competence, transparency
- Inclusion, accountability
-  Remember: Ethics is not just knowing the rules – it's doing the right thing.

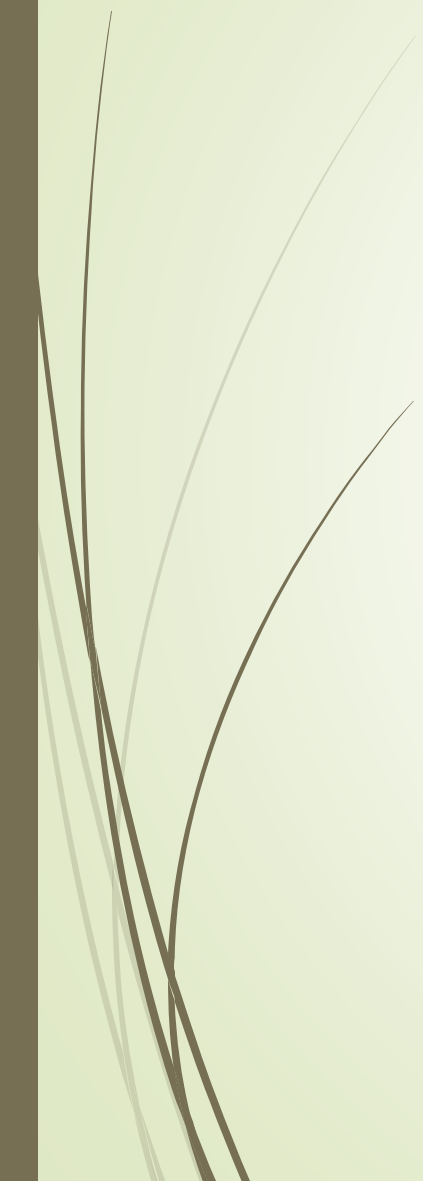


Applications in Real Life

- Ethics in:
 - - Software development
 - - Data privacy
 - - AI & Robotics
 - - Environmental engineering

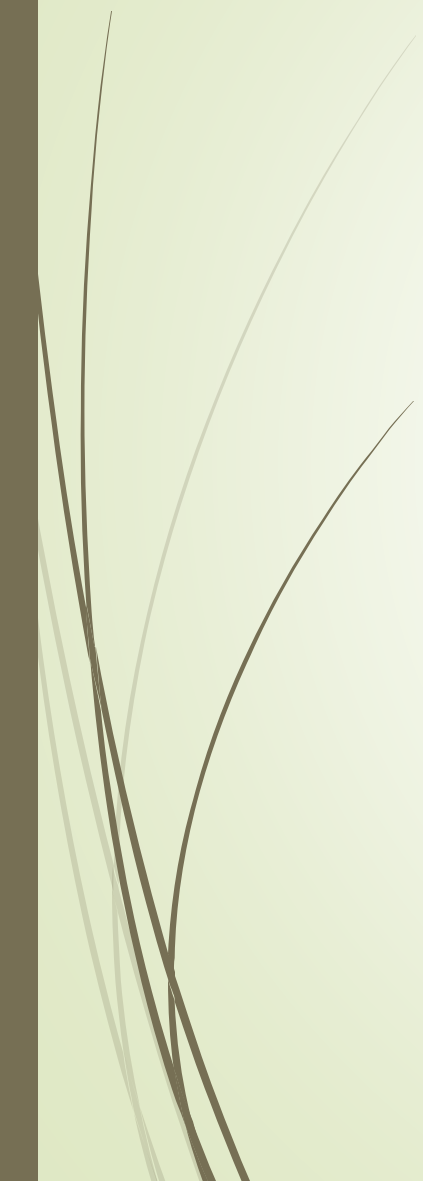


Group Discussion / Activity

- Divide into groups
 - Choose a principle and discuss a real-world ethical dilemma
 - Present findings
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Conclusion

- IEEE Code is a living guide
 - Engineers must lead with ethics
 - Uphold trust in technology
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References

- IEEE Code of Ethics (June 2020)
- www.ieee.org
- Additional ethical guidelines (ACM, NSPE, etc.)